PROJECT IV

Customer Support Ticket Analyser – Project Summary & Insights

Project Overview

The Customer Support Ticket Analyser was developed using Python to analyze customer support tickets and generate meaningful insights. The project demonstrates the use of Python fundamentals such as dictionaries, lists, loops, conditional statements, string manipulation, functions, and sets to clean, organize, and analyze ticket data.

Objectives

- Display customer support ticket information in a readable format.
- Add new customer tickets dynamically with input validation.
- Clean and standardize issue descriptions.
- Analyze tickets based on keywords.
- Generate summary statistics and useful business insights.

Data Cleaning Performed

The issue descriptions were standardized using the following techniques:

- Removed leading and trailing spaces.
- Converted all text to lowercase.
- Removed punctuation marks.
- Reduced multiple spaces to a single space.
- Standardized text formatting for consistent analysis.

Analysis Performed

1. Ticket Management

- Displayed all existing customer tickets.
- Allowed users to add new tickets with auto-generated ticket numbers.
- Validated priority values (High, Medium, Low) before storing data.

2. Keyword Analysis

A reusable Python function was created to count the number of ticket descriptions containing specific keywords. The analysis was performed for the following words:

- Poor
- Good
- Slow
- Excellent

The function performs a case-insensitive search and counts the number of tickets containing the specified keyword.

3. Priority Analysis

The project categorizes tickets based on their priority levels:

- High Priority
- Medium Priority
- Low Priority

This helps identify the distribution of support requests according to urgency.

4. Additional Insights

The project also identifies:

- The ticket with the longest issue description.
- The total number of unique words used across all issue descriptions.
- A sorted list of unique words to better understand frequently used customer terms.

Key Learnings

Through this project, I gained practical experience in:

- Working with dictionaries containing multiple lists.
- Using loops to process structured data.
- Applying string methods for data cleaning.
- Creating reusable Python functions.
- Performing keyword-based searches.
- Validating user input using loops.
- Using sets to identify unique values.
- Generating meaningful insights from textual data.

Conclusion

This project demonstrates how Python can be used to clean, organize, and analyze customer support data effectively. By converting raw ticket information into structured insights, the system helps identify common customer issues, monitor service quality, and support better decision-making. The project also strengthened my understanding of Python programming and data analysis concepts through a real-world use case.